

Identification and Classification of Cross-outs in Handwritten Documents

D7047E – Advanced Deep Learning
Project Report

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1 Abstract

Handwritten text recognition (HTR) is a critical task in document analysis, yet it is often hindered by the presence of cross-outs, where words are marked with lines or scribbles. These artifacts obscure content and significantly affect recognition accuracy. This project develops a deep learning-based approach to automatically identify and classify seven types of synthetic cross-outs applied to the IAM dataset. We propose a multi-stage pipeline involving binary detection and multi-class classification. Furthermore, we evaluate model robustness through a custom-curated dataset of 56 images to ensure generalization across unseen handwriting styles.

2 Introduction

Cross-outs in handwritten documents signify author edits but act as noise for automated recognition systems [1]. This project develops a deep learning pipeline to (1) detect crossed-out words (binary), (2) classify the cross-out style into one of seven types, and (3) evaluate generalisation on a custom dataset of 56 real hand-drawn images [1].

3 Background and Model Selection

Deep learning has become the dominant approach for artifact classification in handwritten documents:

- **CNNs** (ResNet, EfficientNet): learn hierarchical spatial features effective for local stroke patterns.
- **Transfer Learning:** fine-tuning ImageNet pre-trained models achieves high accuracy with limited document data.
- **Vision Transformers (ViT):** capture long-range spatial dependencies via self-attention — useful for patterns like diagonal or zig-zag cross-outs spanning entire words.

4 Method

4.1 Dataset and Structure

The project uses a modified IAM Handwriting Database [2] with seven synthetic cross-out styles: Single-Line, Double-Line, Diagonal, Cross, Wave, Zig-zag, and Scratch — plus a **CLEAN** and **MIXED** class. For binary classification all styles and **MIXED** collapse into one positive class (1:8 imbalance); for multi-class, **MIXED** is excluded leaving eight balanced classes (Table 1).

Table 1: Dataset Splits

Split	CLEAN	CROSSED [†]	MC per class [‡]
Train	47,997	383,976	47,997
Val	7,559	60,472	7,559
Test	20,306	162,448	20,306

[†] 7 styles + MIXED ($8 \times 47,997$ per split).

[‡] Multi-class: 8 balanced classes (MIXED excluded).

4.2 Preprocessing Pipeline and Data Augmentation

Table 2: Preprocessing and Augmentation Pipeline

Transform	Parameters	Purpose
<i>Deterministic (train, val, test)</i>		
Grayscale $\rightarrow$ 3-channel	—	ImageNet compatibility
PadToSquare	fill=255	Preserve stroke aspect ratio
Resize	224×224	Architecture input requirement
Normalize	ImageNet μ/σ	Consistent with pre-trained weights
<i>Stochastic (training only)</i>		
RandomRotation	$\pm 5^\circ$	Tilted paper/camera
RandomAffine	5% translation	Positional shift robustness
RandomPerspective	distortion 0.2, $p=0.5$	Angled photo capture
GaussianBlur	$\sigma \in [0.1, 1.0]$	Camera blur / low-res scanning
ColorJitter	brightness & contrast ± 0.3	Variable lighting

4.3 Implementation Strategy

A two-stage pipeline was implemented: Stage 1 (binary) detects whether a word is crossed out; Stage 2 (multiclass) identifies the style. SimpleCNN (trained from scratch) is the final model for both tasks. ResNet-50 and ViT-B/16 were explored as alternatives. Generalisation is assessed on a custom dataset of 56 real hand-drawn images.

4.4 Binary Classification

4.4.1 Task Definition

CLEAN (label 0) vs. CROSSED-OUT (label 1); all seven styles and MIXED collapsed into the positive class. The 1:8 class imbalance (Table 1) is addressed with class-weighted cross-entropy.

4.4.2 Architecture

The selected model is **SimpleCNN**, a custom CNN trained from scratch with 7 convolutional layers organised into four blocks, all using 3×3 kernels with BatchNorm and ReLU:

- Block 1 (224→112): Conv(3→32) ×2 + MaxPool
- Block 2 (112→56): Conv(32→64) ×2 + MaxPool
- Block 3 (56×56, no pool): Conv(64→128) + Conv(128→128) — preserves spatial detail
- Block 4 (56×56, no pool): Conv(128→256) + AdaptiveAvgPool(7×7)

Classifier head: Flatten(12,544) → Linear(512) → ReLU → Dropout(0.4) → Linear(128) → ReLU → Dropout(0.3) → Linear(2). Total trainable parameters: 7.07M.

We also conducted early-epoch trials with ResNet-50 and ViT-B/16 (ImageNet pre-trained, frozen backbone). After observing that SimpleCNN matched or exceeded their validation accuracy within the first 5–10 epochs while being significantly faster to train, we proceeded exclusively with SimpleCNN for the binary task.

4.4.3 Training Configuration

Table 3 summarises the hyperparameters for all models across both tasks.

Table 3: Training Hyperparameters — All Models

Hyperparameter	Binary (SimpleCNN)	MC (SimpleCNN)	MC (ResNet-50)
Optimizer	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-4}	4×10^{-4}
LR scheduler	ReduceLROnPlateau (factor=0.1, patience=5, min_lr= 10^{-6})		
Batch size	64	64	256
Max / Min epochs		100 / 20	
Early stopping		15 epochs	
Loss function	Wtd. CE (1:8)	Cross-Entropy (balanced classes)	
Unfrozen layers	All	All	layer4 + fc head

Regularization. Dropout ($p=0.4$, $p=0.3$) and BatchNorm in the classifier head; class-weighted loss (factor 8) for the 1:8 imbalance; ReduceLROnPlateau and early stopping as per Table 3; data augmentation (Section 4.2) as the primary input-space regularizer.

4.4.4 Results

Table 4 summarises performance on the IAM test set and the custom evaluation set.

Table 4: Binary Classification — SimpleCNN Results

Metric	IAM Test Set	Custom Dataset
F1 Score	0.9908	0.9320
Generalisation Gap		0.0587

SimpleCNN achieved IAM F1 = 0.991, converging within ~ 5 epochs, and generalises well to the custom evaluation set (F1 = 0.932, gap = 0.059). ResNet-50 and ViT-B/16

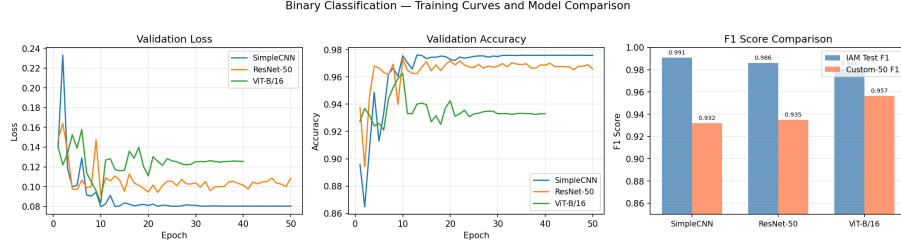


Figure 1: Binary classification training curves: SimpleCNN, ResNet-50, and ViT-B/16.

were trialled but discontinued after 5–10 epochs as SimpleCNN matched their validation accuracy at significantly lower training cost.

4.5 Multi-class Classification

4.5.1 Task Definition

Eight balanced classes: **CLEAN** plus the seven cross-out styles. **MIXED** is excluded as it has no specific style label (see Table 1).

4.5.2 Architectures

Two architectures were trained and compared:

SimpleCNN. The same architecture described in Section 4.4, with the output layer widened to 8 classes. Trained from scratch with $LR = 10^{-4}$, batch size 64.

ResNet-50. Pre-trained on ImageNet. Unlike the binary variant where the entire backbone was frozen, here **layer4** (the final residual block, $\sim 8.4M$ parameters) is unfrozen alongside the classification head to allow fine-grained feature adaptation. The head is: $\text{Linear}(2048 \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(512 \rightarrow 8)$. Trained with $LR = 4 \times 10^{-4}$, batch size 256.

4.5.3 Training Configuration

Hyperparameters follow Table 3. Both models use the same ReduceLROnPlateau scheduler, early stopping, and epoch budget; they differ in learning rate, batch size, and unfrozen layers.

4.5.4 Results

IAM Test Set. Table 5 summarises IAM test set performance for both models.

Table 5: Multi-class Classification — IAM Test Set Results

Model	Val Acc	Test Acc	Macro-F1	Custom Dataset F1
ResNet-50	0.8947	0.9025	0.9098	—
SimpleCNN	0.8957	0.9042	0.9130	0.485

Training dynamics. Both models converged to ≈ 0.895 validation accuracy (ResNet-50 in 55 epochs, SimpleCNN in 77). Despite training longer, SimpleCNN achieved marginally higher test accuracy (0.9042 vs. 0.9025) and Macro-F1 (0.9130 vs. 0.9098).

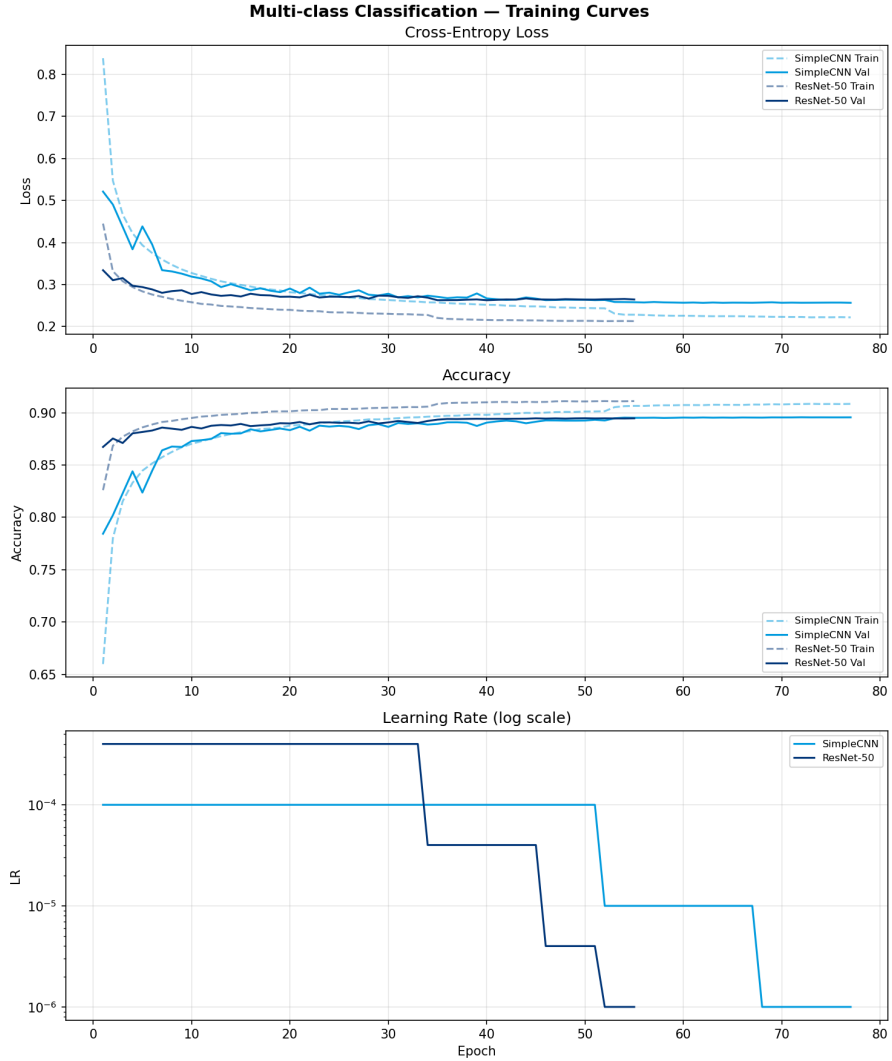
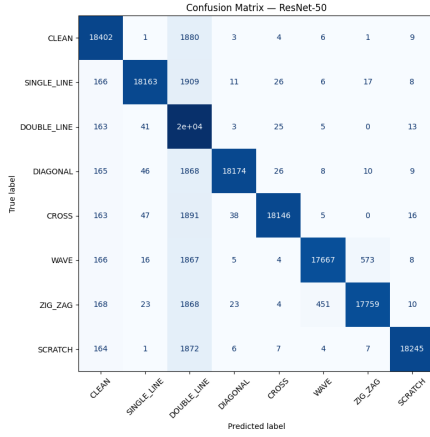
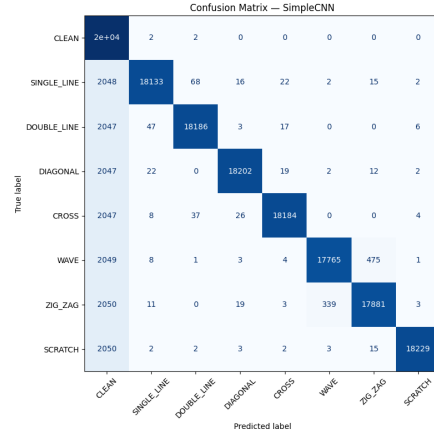


Figure 2: Multi-class training curves: ResNet-50 vs. SimpleCNN.

Confusion matrix analysis. Both models show systematic **ZIG ZAG over-prediction** ($\approx 9\text{--}10\%$ of every class), with WAVE most affected in ResNet-50 (12.9%). SimpleCNN’s pattern is slightly attenuated, consistent with its higher Macro-F1 (Figure 3).



(a) ResNet-50



(b) SimpleCNN

Figure 3: Confusion matrices on the IAM test set. SGL = Single-Line, DBL = Double-Line, DIAG = Diagonal, ZZ = Zig-Zag.

Model selection. SimpleCNN is selected as the final multiclass model:

- Higher Macro-F1 (0.9130 vs. 0.9098) and test accuracy (0.9042 vs. 0.9025)
- Lighter: 7.07M vs. ~ 25 M parameters
- ResNet-50’s DOUBLE_LINE over-prediction is an ImageNet pretraining artifact requiring costly deeper unfreezing
- Consistent with the binary task choice

Custom Dataset Generalisation. SimpleCNN drops to Macro-F1 = 0.485 (gap = 0.428 vs. IAM), per Table 6 and Figure 4. Three failure modes emerge:

- **WAVE & ZIG_ZAG $\rightarrow$ CLEAN:** real strokes are lighter and less uniform than synthetic training examples.
- **SINGLE_LINE, DOUBLE LINE & CROSS $\rightarrow$ DIAGONAL:** natural hand-drawn lines carry a slight angle vs. perfectly horizontal synthetic strokes.
- **CLEAN & SCRATCH robust** (F1 = 1.0 each): unambiguous appearance survives the domain shift.

Root cause: the **synthetic-to-real domain gap** — clean 300 dpi IAM scans vs. hand-photographed images with variable lighting, stroke pressure, and perspective.

Table 6: Custom Dataset Per-Class Results (SimpleCNN)

Class	P	R	F1
CLEAN	0.33	1.00	0.50
SINGLE.LINE	0.50	0.14	0.22
DOUBLE.LINE	0.40	0.29	0.33
DIAGONAL	0.55	0.86	0.67
CROSS	0.75	0.43	0.55
WAVE	1.00	0.14	0.25
ZIG.ZAG	1.00	0.29	0.44
SCRATCH	1.00	1.00	1.00
Macro avg	0.69	0.52	0.49

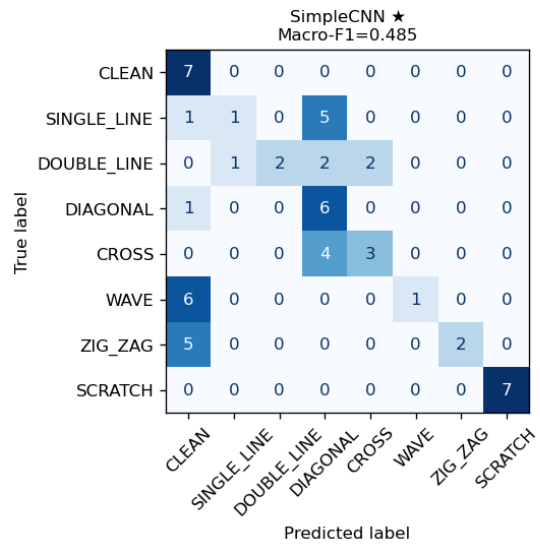


Figure 4: Confusion matrix on the custom dataset (Macro-F1 = 0.485).



Figure 5: All 27 misclassified examples: WAVE/ZIG_ZAG $\rightarrow$ CLEAN (faint strokes); SINGLE_LINE, DOUBLE_LINE, CROSS $\rightarrow$ DIAGONAL (natural hand-drawn angle).

5 Deliverables

1. **Source Code and Trained Models:** Codebase and model weights on GitHub [3], including training notebooks and evaluation scripts.
2. **Custom Evaluation Dataset:** 56 real hand-drawn images ($8 \text{ classes} \times 7 \text{ words}$), available at https://github.com/chirag-gurumurthy/D7047E_ADL/tree/project/project/dataset/custom_50.
3. **Final Presentation:** 10-minute presentation covering the problem, pipeline, results, and domain shift analysis (https://github.com/chirag-gurumurthy/D7047E_ADL/tree/project/project/docs/ADL_Group14_Presentation.pptx).

References

- [1] Pathirage, G. and Barney Smith, E. *ADL Project: Document Analysis*.
- [2] Marti, U. and Bunke, H. *The IAM-database: an English sentence database for offline handwriting recognition*.
- [3] Gurumurthy, C. *D7047E_ADL Project Repository*. https://github.com/chirag-gurumurthy/D7047E_ADL/tree/project/project.